
SMART HOSPITAL APPOINTMENT AND PATIENT PRIORITIZATION SYSTEM

Activity 1: Problem Statement

Project Title

Smart Hospital Appointment and Patient Prioritization System

1. What Problem Is Being Solved?

Many hospitals experience long waiting times and overcrowding because patients are typically processed using a first-come, first-served approach. While this method is simple, it does not always consider the urgency of a patient's medical condition. As a result, patients with minor concerns may be attended to before patients who require more urgent medical attention.

Another challenge is that many patients visit hospitals for services that do not necessarily require consultation with a doctor. Examples include medication collection, routine check-ups, follow-up visits, and basic health assessments that could be handled by nurses, pharmacists, or administrative personnel.

These issues often result in:

- Long queues and waiting periods
- Overworked healthcare professionals
- Delays in treating urgent cases
- Poor patient experience
- Inefficient allocation of hospital resources
- Unnecessary hospital visits

The project aims to improve fairness, efficiency, and accessibility within hospital appointment management.

2. Why Does the Problem Require a Technological Solution?

A technological solution can assist hospitals by assessing patient needs before appointments are booked. Through a structured symptom assessment questionnaire, the system can gather relevant information about a patient's condition and determine the most appropriate healthcare service.

The solution can:

- **Pre-screen patients before arrival**
- **Assess appointment urgency**
- **Reduce unnecessary doctor consultations**
- **Direct patients to the correct healthcare professional**
- **Improve queue management**
- **Support hospital staff with informed scheduling decisions**
- **Reduce waiting times and congestion**

Without technology, these assessments are performed manually by healthcare workers, increasing workload and reducing operational efficiency.

3. Who Are the Target Users?

Primary Users

- **Patients**
- **Elderly patients**
- **Parents booking appointments for children**
- **Family members scheduling appointments for dependents**

Secondary Users

- **Doctors**
 - **Nurses**
 - **Pharmacists**
 - **Reception staff**
 - **Hospital administrators**
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Activity 2: Proposed Solution

System Name

SmartCare Hospital Appointment and Prioritization System

Purpose

The SmartCare Hospital Appointment and Prioritization System is a web-based healthcare management platform designed to improve appointment scheduling and patient prioritization.

Instead of simply allowing patients to select an appointment date, the system first evaluates the patient's healthcare needs using a symptom assessment questionnaire. Based on the responses provided, the system recommends the most appropriate service and assists healthcare personnel in managing appointments more effectively.

The system does not replace healthcare professionals. Instead, it supports existing hospital processes by improving patient screening, appointment allocation, and queue management.

Key Features

Patient Registration and Login

Patients can:

- **Create accounts**
- **Manage personal information**
- **View appointment history**
- **Track appointment status**

Symptom Assessment Questionnaire

Before scheduling an appointment, patients complete a questionnaire related to:

- **Fever**
- **Headaches**
- **Chest pain**
- **Medication collection**
- **Follow-up consultations**
- **Injuries**
- **Chronic illnesses**

The questionnaire helps determine the most appropriate healthcare service for the patient.

Intelligent Patient Prioritization

The system calculates a priority score based on factors such as:

- **Reported symptoms**
- **Age**
- **Existing medical conditions**
- **Pregnancy status**
- **Disability status**

Example priority levels:

- **High Priority**
- **Medium Priority**
- **Standard Priority**

The final decision remains subject to review by healthcare personnel.

Smart Appointment Scheduling

The system recommends appropriate appointments for:

- **Doctors**
- **Nurses**
- **Pharmacists**
- **Follow-up consultations**

This helps reduce unnecessary doctor visits and improves resource utilization.

Digital Queue Management

The system generates queue positions based on:

- **Appointment time**
- **Patient priority level**
- **Patient category**

This supports fairer and more efficient patient processing.

Notification Services

The platform provides notifications through:

- Email
- SMS

Notifications include:

- Appointment confirmations
- Appointment reminders
- Queue updates
- Schedule changes

Healthcare Staff Dashboard

Healthcare personnel can:

- View patient appointments
- Review assessment results
- Manage schedules
- Update appointment statuses
- Monitor daily patient queues

Expected Output

The system is expected to produce:

- Prioritized appointment schedules
 - Reduced waiting times
 - Better patient distribution across healthcare services
 - Improved queue management
 - Enhanced patient experience
 - Improved hospital operational efficiency
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Activity 3: System Design

System Architecture

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| Patients |

+-----+-----+

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| Web Application |

| Interface |

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| Authentication Layer |

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| Symptom Assessment |

| Module |

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| Patient Priority |

| Evaluation Engine |

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| Appointment |
| Scheduling Module |
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| Staff || Notification|
| Dashboard || Service |
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| PostgreSQL Database |
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Core Database Entities

The database will initially include:

Users

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+-- Patients
|

+-- Staff

Patients

|

+-- Appointments

|

+-- Assessments

Assessments

|

+-- AssessmentAnswers

Appointments

|

+-- QueueEntries

Staff

|

+-- Schedules

Notifications

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Activity 4: Technology Selection

Component	Technology
Frontend	React.js
Backend	Java Spring Boot

Component	Technology
Database	PostgreSQL
Authentication	Spring Security + JWT
API Development	REST API
Email Notifications	Spring Mail
SMS Integration	Twilio API
Hosting	Azure / AWS / Render
Version Control	GitHub
Development Tools	IntelliJ IDEA, VS Code, Postman

Intelligent Assessment Engine

The first version will use a rule-based assessment system rather than Artificial Intelligence.

Example:

If:

Chest Pain = Yes

Difficulty Breathing = Yes

Then:

Priority = High

If:

Medication Collection

Then:

Service = Pharmacy

If:

Blood Pressure Check

Then:

Service = Nurse

Future versions may incorporate machine learning models trained on historical healthcare data to improve prioritization accuracy.

Activity 5: Prototype / Demonstration

Screen 1: Patient Login

SMARTCARE LOGIN

Email Address

Password

[Login]

[Register]

Screen 2: Symptom Assessment

APPOINTMENT REQUEST

Select Reason for Visit

- ☐ Fever
- ☐ Headache
- ☐ Chest Pain
- ☐ Medication Collection
- ☐ Follow-Up Consultation
- ☐ Injury

[Next]

Screen 3: Assessment Result

ASSESSMENT RESULT

Priority Level:

Medium Priority

Recommended Service:

Doctor Consultation

Available Date:

12 September 2026

Available Time:

09:30 AM

[Confirm Appointment]

Screen 4: Staff Dashboard

TODAY'S APPOINTMENTS

Patient Name Priority

John Doe High

Mary Smith Medium

James Brown Standard

[View Details]

Project Innovation

Most hospital appointment systems focus solely on booking dates and times.

The SmartCare Hospital Appointment and Prioritization System introduces an intelligent pre-screening process that evaluates patient needs before appointments are scheduled.

The system innovates by:

- **Assessing patients before hospital visits**
- **Supporting fairer appointment allocation**
- **Prioritizing potentially urgent cases**
- **Reducing unnecessary doctor consultations**
- **Improving healthcare resource utilization**

- **Reducing waiting times and queue congestion**

Rather than replacing existing hospital procedures, the system enhances them through technology, creating a more efficient and patient-centered healthcare experience.

"The right patient, at the right service, at the right time."

It's short, professional, and immediately tells examiners what problem the system is solving.